INCH-POUND
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SUPERSEDING
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November 9, 1990

COMMERCIAL ITEM DESCRIPTION

BLOCKS, TACKLE, WIRE ROPE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers single- and multiple-sheave tackle blocks, snatch blocks, and cargo hoisting blocks for use with wire rope.

2. CLASSIFICATION. The blocks shall be of the following types, classes, sizes, and rigs as specified (see 7.2).

2.1 Types and classes. The types and classes of blocks are as follows:

a. Type I - Blocks, oval pattern (see figure 1).
   (1) Class 1 - Single sheave.
   (2) Class 2 - Double sheave.
   (3) Class 3 - Triple sheave.

b. Type II - Blocks, diamond pattern (see figure 2).
   (1) Class 1 - Single sheave.
   (2) Class 2 - Double sheave.
   (3) Class 3 - Triple sheave.

c. Type III - Blocks; snatch, safety locking (see figure 3).

d. Type IV - Blocks, cargo hoisting (see figure 4).

2.2 Sizes. The sizes of the blocks are as specified in tables I through IV.

2.3 Rigs. The rigs for the blocks are as depicted on figure 5.

3. SALIENT CHARACTERISTICS.

3.1 Design and construction. Block design shall be as shown in tables I through IV and on figures 1 through 5, as to type, class, size, and rig. Each block shall be free from any defects that would adversely affect the performance or maintainability of individual components. Replacement parts shall be manufactured to standard tolerances to permit replacement without requiring modification. All parts subject to wear shall be removable without dismantling adjacent parts.

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FSC 3940

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TABLE I. Type I blocks, oval pattern.

Size	Wire diameter (inches)	Sheave (inches)			Working load (pounds)		
		Outside diameter	Rim width	Center pin	Single	Double	Triple
6	⅜	6	1	¾	4,000	5,000	6,000
8	½	8	1¼	⅞	5,000	7,000	9,000
10	⅝	10	1¼	⅞	6,000	9,000	11,000
12	⅝	12	1½	1	8,000	12,000	14,000
14	¾	14	1½	1⅛	10,000	14,000	16,000
16	⅞ - 1	16	1¾	1½	16,000	30,000	36,000
18	1	18	1¾	1½	20,000	36,000	40,000

TABLE II. Type II blocks, diamond pattern.

Size	Wire diameter (inches)	Sheave (inches)			Working load (pounds)		
		Outside diameter	Rim thickness	Center pin	Single	Double	Triple
6	⅜	6	1	¾	4,000	5,000	6,000
8	½	8	1¼	⅞	5,000	7,000	9,000
10	⅝	10	1¼	⅞	6,000	9,000	11,000
12	⅝	12	1½	1	8,000	12,000	14,000
14	¾	14	1½	1⅛	10,000	14,000	16,000
16	⅞ - 1	16	1¾	1½	16,000	30,000	36,000
18	1	18	1¾	1½	20,000	36,000	40,000

TABLE III. Type III blocks; snatch, safety locking.

Size	Wire diameter (inches)	Sheave (inches)			Working load (pounds)
		Outside diameter	Rim thickness	Center pin	
4	⅜	4	¾	½	2,000
6	⅜ - ½	6	1	¾	4,000
8	½ - ⅝	8	1¼	⅞	6,000
10	⅝	10	1¼	1	8,000
12	¾	12	1½	1⅛	10,000
14	¾ - ⅞	14	1½	1¼	12,000
16	⅞ - 1	16	1¾	1½	15,000
18	1	18	1¾	1½	20,000
20	1¼	20	2¼	1¾	24,000
20	1½	20	2¼	1¾	28,000

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TABLE IV. Type IV blocks, cargo hoisting.

Size	Wire diameter (inches)	Sheave (inches)		Working load (pounds)
		Outside diameter	Rim thickness	
12	¾	12	1¾	10,000
14	¾ - ⅞	14	1¾	16,000
16	¾ - 1¼	16	---	20,000
18	¼ - 1¼	18	---	20,000

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the manufacturer and shall be subject to all provisions of this CID. Material shall be new and unused.

3.2.1 Castings. Castings shall be of uniform quality and free of blowholes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects. Strength, heat treatment, chemical composition, tensile properties, and other essential physical properties of the castings specified (see 7.2) shall be in accordance with ASTM A27/A27M for carbon steel and ASTM A148/A148M for carbon and low alloy steel.

3.2.2 Forgings. Forgings shall be of uniform quality, free of flash, scale, cracks, hard spots, cold shuts, and excessive cooling stresses. Strength and other essential physical properties of the forgings specified (see 7.2) shall be in accordance with ASTM A668/A668M.

3.2.3 Welding. Welding, if necessary, shall be performed in accordance with S9074-AR-GIB-010/278. Welding shall not be used to repair blocks.

3.2.4 Fastening devices. Screws, pins, bolts, and similar parts subject to removal, tightening, or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.2.5 Threads. Threaded parts shall be in accordance with the applicable requirements of FED-STD-H28.

3.2.6 Surface. Castings, forgings, stampings, and welded parts shall be thoroughly cleaned and free of sand, dirt, scale, flux, or other harmful or extraneous materials. External surfaces shall be free from burrs, sharp edges, and corners. All bearing surfaces between matching parts shall be finished to the necessary tolerances to provide the required stability in operation. Surface roughness of bearing surfaces shall not reduce bearing areas appreciably or detract from appearance.

3.2.7 Galvanizing. Metal surfaces of blocks except bearing surfaces and bushings shall be hot dip galvanized in accordance with ASTM A153/A153M, class B3. The basic metal shall be clean and free from defects that will impair appearance or protective value of the coating. Galvanizing shall be done prior to assembly of component parts.

3.2.8 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this CID shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered material” means waste materials and by-products recovered or diverted from solid waste, but the term does not include those materials and by-products generated from, and commonly reused within, an original manufacturing process. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this CID.

3.3 Rig assemblies. Blocks and fittings of corresponding size and strength shall be assembled into rigs and identified by rig numbers (see 3.4 and figure 5). Special blocks and rigs not designated on figure 5 shall be assembled as specified (see 7.2).

3.4 Identification markings. Each block assembly shall be permanently and legibly marked by stamping the shell plate with the manufacturer’s name or trademark, the safe working load (SWL), and the size of the block, which shall be listed as the outside diameter of the sheave to conform with current commercial practice (see tables I through IV). Abbreviations may be used. Identification marking shall be in accordance with MIL-STD-130.

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3.5 Details of type I, II, III, and IV blocks.

3.5.1 Block components.

3.5.1.1 Shells. Shell plates shall be made of steel casting, structural steel, or carbon steel hot-rolled sheet, as required, in accordance with ASTM A27/A27M, ASTM A36/A36M, or ASTM A159, respectively. The edges shall be well rounded.

3.5.1.2 Straps. Steel straps shall be made of structural steel, cold-rolled carbon steel, or hot-rolled carbon steel, as required, in accordance with ASTM A36/A36M, ASTM A109/A109M, or ASTM A1011/A1011M, respectively. Steel straps shall support the center pin and transmit sheave loads to the fitting (hook, shackle, swivel, etc.), as required.

3.5.1.3 Sheaves. Unless otherwise specified (see 7.2), sheaves shall be cast steel in accordance with ASTM A27/A27M or forged steel in accordance with ASTM A668/A668M, and shall be designed to prevent cutting action on the rope. The rim and tread section of each sheave shall be properly grooved to fit the contour of the rope under load conditions. The sheaves shall be concentric with the bore, and shall be designed so that only the hub bears against the shell. Side play shall not exceed ⅛ inch. Sheaves shall have a smooth-running action. Sheaves shall conform to the minimum dimensions listed in tables I through IV.

3.5.1.4 Bushings. Each sheave shall be furnished with a one-piece bronze bushing in accordance with ASTM B438, grade 1, class A, type II, which shall be pressed into the sheave with sufficient tightness to prevent slippage under extreme loads. The bushing shall be equal in length to the sheave thickness at the hub and shall have a minimum wall thickness of $\frac{3}{16}$ inch.

3.5.1.5 Roller bearings. Bearings shall be tapered roller bearings, which can take both load and side thrust and hold the sheave centrally so it cannot chafe or wear on the sides. The bearings shall be locked in a grease-packed chamber with felt washers to retain the grease and exclude dirt.

3.5.1.6 Center pin. The center pin shall be of carbon steel in accordance with ASTM A108, having a carbon content of 0.25 to 0.50 percent. The pin shall transmit the sheave load to the shell or straps without rotating (see tables I, II, and III for minimum pin diameters).

3.5.1.7 Fittings. Each block shall include a shackle, swivel shackle, hook, or swivel hook, as specified (see 7.2), using the applicable rig number shown on figure 5 to identify the type of fitting required. Safety factors of fittings shall be as specified in 3.6.1.

3.5.1.8 Hook. The hook shall be drop forged from carbon steel, with heavier sections at the areas which take the maximum stress. Hook safety factors shall be as specified in 3.6.1.

3.5.1.9 Becket. The becket shall be formed by the extension of the strap ends, and shall show no distortion with the wire rope dead ended at the becket and the loads specified in tables I and II suspended from the hook.

3.5.1.10 Spacers and roller guides. Spacers shall be provided as required to make the block a strong, rugged unit with the required sheave clearances. Roller guides shall be provided as required to prevent the wire from coming out of the sheaves when the block is overhauled.

3.5.2 Type I blocks, oval pattern. Type I blocks shall be in accordance with 3.1, figure 1, and table I and as specified in 3.5.1.1 through 3.5.1.4 and 3.5.1.6 through 3.5.1.10.

3.5.3 Type II blocks, diamond pattern. Type II blocks shall be in accordance with 3.1, figure 2, and table II and as specified in 3.5.1.1 through 3.5.1.4 and 3.5.1.6 through 3.5.1.10.

3.5.4 Type III blocks; snatch, safety locking. Type III blocks shall be in accordance with 3.1, figure 3, and table III and as specified in 3.5.1.3, 3.5.1.4, 3.5.1.6, 3.5.1.10, and 3.5.4.1 through 3.5.4.3.

3.5.4.1 Shell. The shell shall be as specified in 3.5.1.1 and shall have a rope guard indented around the top of the shell to prevent jamming of the rope between the sheave and the shell.

3.5.4.2 Fittings. Blocks shall be fitted with either a swivel hook or a swivel eye with shackle as specified (see 7.2).

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3.5.4.3 Safety locking device. Each block shall be provided with a self-locking safety device that will hold the block securely closed under load as specified (see 7.2).

3.5.5 Type IV blocks, cargo hoisting. Type IV blocks shall be in accordance with 3.1, table IV, and figure 4 and as specified in 3.5.1.3, 3.5.1.5, and 3.5.5.1 through 3.5.5.3.

3.5.5.1 Shell. Shells shall be as specified in 3.5.1.1 with recessed mortise.

3.5.5.2 Center pin. The center pin shall be as specified in 3.5.1.6 and shall contain a recessed grease fitting in the head and grease grooves.

3.5.5.3 Fittings. Blocks shall be rigged as rig 1 and rig 3; in addition, they may be self-adjusters shown as rig 15. All swivel connections shall contain a recessed Alemite grease fitting, or equal, and all fittings shall be in accordance with 3.5.1.7.

3.6 Physical requirements.

3.6.1 Safe working load (SWL). The SWL shall be based on a safety factor of 5 on the minimum breaking strength of the complete block assembly. The SWL, rope, and sheave size shall be as shown in tables I through IV as applicable for the type block specified.

3.6.2 Proof load. When subjected to a proof load, each block assembly sampled as specified (see 7.2) shall be proof tested to twice the SWL specified in tables I through IV. The applicable proof load shall be applied to the block assembly and held for a period of not less than 10 minutes. At the conclusion of the test, the block and its components shall be carefully examined and measured for signs of fracture, permanent set, or evidence of instability, deformation, distortion, cracks, excessive wear, or mechanical defects.

3.6.3 Breaking load. Each block assembly sampled as specified (see 7.2) shall be tested for breaking strength to destruction. A steadily increasing load shall be applied until failure occurs. When tested to destruction, the block or fittings shall not fail at less than five times the SWL specified in tables I through IV.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The manufacturer must have at least 5 years of production experience and must have produced at least 50 items meeting the requirements of this CID.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.1.2 Defense and federal standards. Defense and federal standards are available online at http://quicksearch.dla.mil.

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7.1.3 Naval Sea Systems Command (NAVSEA) Publications. NAVSEA Publications are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. NAVSEA Publications are available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.

7.1.4 FAR. The Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Type, class, size, and rig (see 2).
c. Other essential physical properties for castings (see 3.2.1) and forgings (see 3.2.2).
d. Assembly of special blocks and rigs not designated on figure 5 (see 3.3).
e. Sheave material and design (see 3.5.1.3).
f. Fittings required (see 3.5.1.7 and 3.5.4.2).
g. Locking connections required (see 3.5.4.3).
h. Block assembly sampling requirements (see 3.6.2 and 3.6.3).
i. Product conformance provisions.
j. Packaging requirements.
k. Quantity required.

7.3 Key words.

Becket
Bushing
Hook
Rig
Sheave
Shell
Strap

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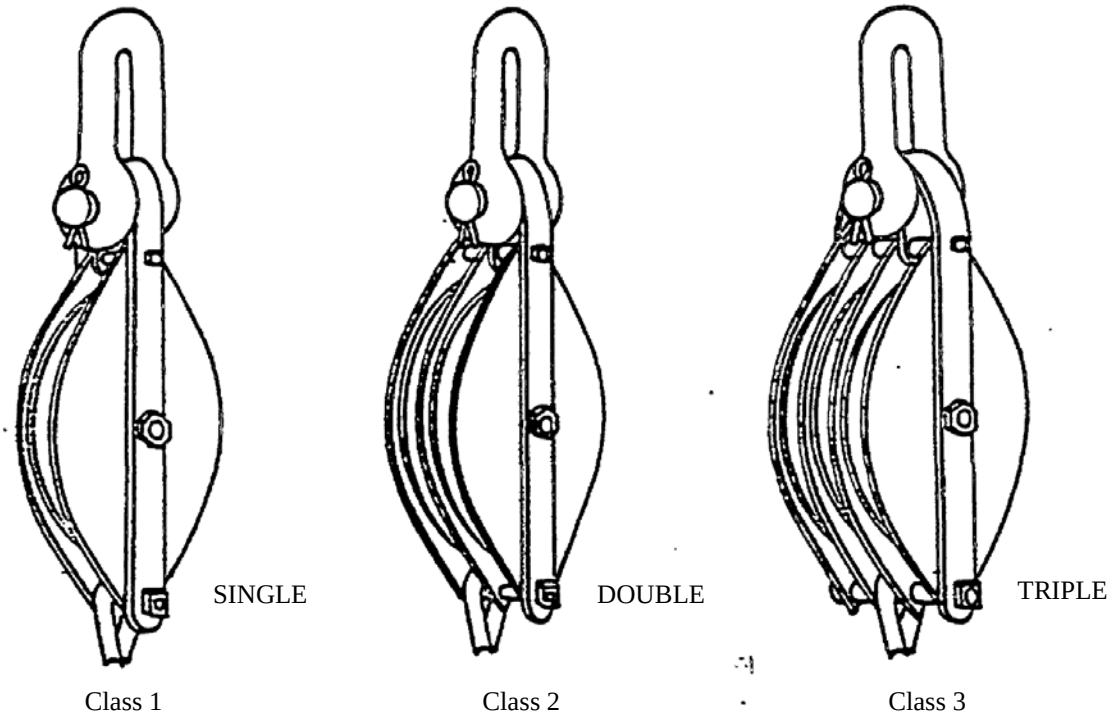


FIGURE 1. Regular pattern oval blocks.

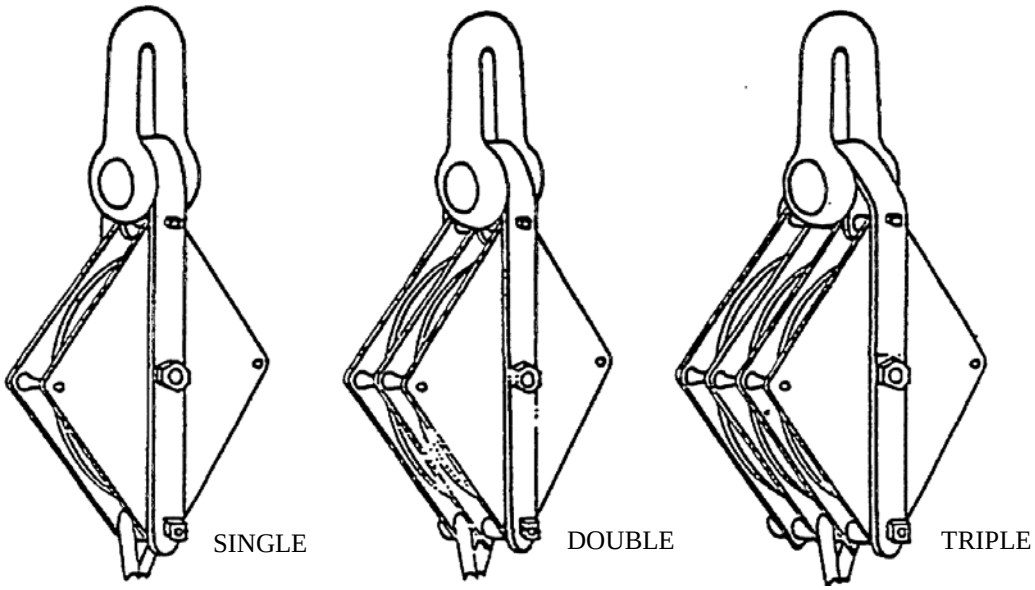
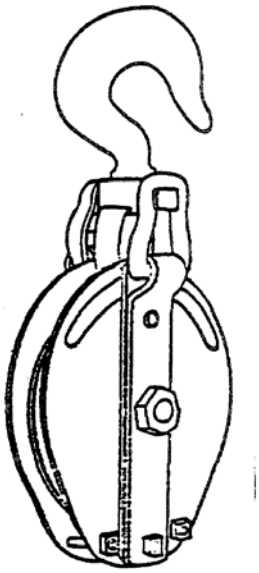


FIGURE 2. Regular pattern diamond blocks.

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SAFETY LOCKING
SWIVEL HOOK

FIGURE 3. Snatch blocks.

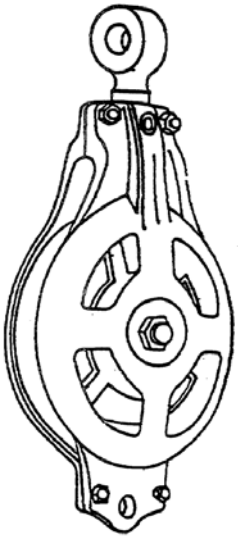


FIGURE 4. Cargo hoisters.

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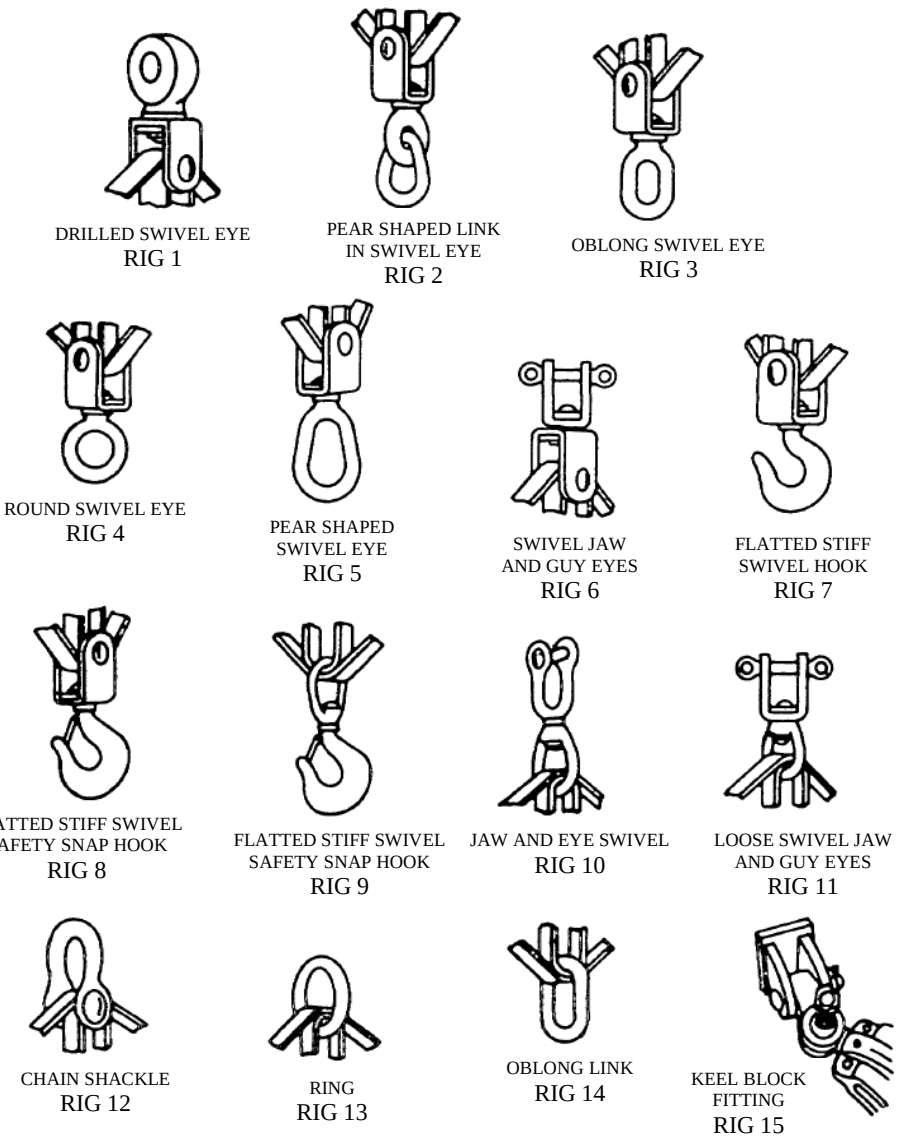


FIGURE 5. Wire rope block rigs.

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MILITARY INTERESTS:

Custodians:
Army – MI
Navy – SH

Review Activities:
Army – AV
DLA – IS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
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